

PAPER The Medical-Isotope Mission of a Compact
Spherical-Tokamak Breeder: Mo-99, Targeted-Alpha Emitters,
and Byproduct Isotope Co-Production on a 14 MeV Neutron
Source

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Abstract

A tritium breeder is, before it is anything else, a bright and steady source of 14 MeV neutrons, and that source has value beyond breeding. We formalise the isotope-production and materials-qualification missions of the compact spherical-tokamak breeder (Hyperion) at its frozen operating point — fusion power $P_{\text{fus}} = 85.04$ MW, gain $Q = 3.076$, plasma current $I_p = 9.66$ MA, negative triangularity $\delta = -0.30$ — which sustains a whole-device source strength $S_n = 3.009 \times 10^{19} \text{ n s}^{-1}$ and a first-wall fast flux of $7.0 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$. We derive the governing production and depletion equations, the two-step Bateman chain of the molybdenum-99/technetium-99m generator, the spectrum-folded reaction-rate integral, and a formal materials-qualification figure of merit, and we cast the source-strength/centrepost-damage coupling as a strict proportional constraint. On this basis a first-wall $^{100}\text{Mo}(n, 2n)^{99}\text{Mo}$ target with no moderator and no added hardware reaches a frozen end-of-bombardment activity of $\approx 1,981$ Ci per 100 g of natural molybdenum after a 14-day irradiation (near its 2,041 Ci saturation), a fission-free route to a diagnostic supply chain that periodically fails. The same field provides a genuine fusion-spectrum irradiation platform: a full-rate fast-neutron capability about $59\times$ a reference dedicated materials facility, and a $\geq 10 \text{ dpa fpy}^{-1}$ test volume of 21.6 L — $315\times$ the IFMIF and $630\times$ the IFMIF-DONES equivalent — in a true fusion spectrum rather than an accelerator surrogate. Targeted-alpha ($^{225}\text{Ac}/^{211}\text{At}$) and byproduct-isotope routes are scoped as capability projections. All results were computed by an NVIDIA GPU high-performance-computing campaign (OpenMC/ENDF-B-VIII.0 transport, FISPACT-II activation, SPECTRA-PKA→LAMMPS damage) and every headline number is a frozen anchor in the Kronos de-risking register. These are strategic-material co-products of the neutron economy, booked as capability, not revenue.

Keywords: medical isotopes; molybdenum-99; technetium-99m; targeted alpha therapy; 14 MeV neutrons; materials qualification; strategic materials; spherical-tokamak breeder

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1 Introduction

1.1 The neutron a breeder makes has more than one use

A deuterium–tritium reaction releases 17.59 MeV, of which 14.06 MeV leaves the plasma as a fast neutron. In a tritium breeder that neutron is not a waste stream to be shielded against but the working fluid of the entire device: it multiplies and is captured on lithium to regenerate the fuel, it displaces atoms in every surrounding component, and — if a target is placed in its path — it transmutes stable nuclei into radioisotopes. The compact spherical-tokamak (ST) breeder Hyperion is designed around this fact. It is a strategic-materials platform rather than a power plant: judged as a generator it would fail a category test it was never built to take, because its products are tritium, helium-3, and neutrons, and its figures of merit are the breeding ratio and the isotope yield, not the megawatt [2, 3]. This paper scopes two of those neutron-economy co-products from the frozen source: the *medical- and byproduct-isotope* mission, built around the molybdenum-99/technetium-99m diagnostic chain, and the *materials-qualification* mission, a true fusion-spectrum irradiation volume for the fusion-materials programme.

1.2 Two shortage-prone supply chains that a fusion source addresses

Medical isotopes. Technetium-99m (^{99m}Tc , $t_{1/2} = 6.0\text{ h}$) is the workhorse of nuclear medicine, used in roughly four of every five diagnostic scans, and it is supplied through the decay of its parent molybdenum-99 (^{99}Mo , $t_{1/2} = 65.9\text{ h}$) [10, 11]. Essentially the entire world supply of ^{99}Mo is produced by fission of highly enriched or low-enriched uranium in a small, ageing fleet of research reactors; the fleet has repeatedly entered simultaneous outage, and each such event propagates into cancelled or delayed patient studies [10, 12, 13]. Because ^{99}Mo cannot be stockpiled — a half-life of under three days consumes any inventory — the supply is structurally fragile. Alternative, non-fission routes have therefore been pursued for two decades, chiefly the neutron routes $^{98}\text{Mo}(n, \gamma)^{99}\text{Mo}$ and $^{100}\text{Mo}(n, 2n)^{99}\text{Mo}$ and the photonuclear route $^{100}\text{Mo}(\gamma, n)^{99}\text{Mo}$ [14, 15]. The $(n, 2n)$ channel has a cross-section near 1.5 b over 12–17 MeV [14], which makes a 14 MeV fusion neutron field an almost ideal driver, and it uses no fissile target, removing the proliferation and high-level-waste burdens of the uranium route [6].

Targeted alpha therapy. A second, faster-growing demand is for alpha-emitting radionuclides for targeted alpha therapy (TAT), where the short range and high linear-energy-transfer of an alpha particle deposits a lethal dose within a few cell diameters of a molecular target. Actinium-225 (^{225}Ac) and its daughter bismuth-213, and astatine-211 (^{211}At), are the leading candidates; the clinical results for ^{225}Ac -PSMA-617 in metastatic castration-resistant prostate cancer transformed the field, but supply is severely constrained [16, 17]. Neutron routes to ^{225}Ac — for example $^{226}\text{Ra}(n, 2n)^{225}\text{Ra} \rightarrow ^{225}\text{Ac}$ — are among the paths under evaluation to break that constraint.

The 14 MeV materials gap. Separately, the fusion-materials programme has never possessed a high-flux, high-volume, true-fusion-spectrum irradiation source. Fission reactors deliver flux and volume but the wrong spectrum (no 14 MeV peak, hence none of the helium- and hydrogen-generating threshold reactions that embrittle a fusion first wall) [21, 20]. Accelerator-based deuteron–lithium sources — IFMIF and its single-beam demonstrator IFMIF-DONES — reproduce the spectrum but only over a small high-flux volume ($\sim 0.5\text{ L}$) and remain in construction [18, 19]. A power-class fusion device is the only object that is simultaneously a large volume and a true fusion spectrum.

1.3 Contribution

We make the isotope and materials missions quantitative and reproducible from a single frozen operating point. Specifically we (i) write the source term, the neutron balance, and the production/depletion equations as an explicit boundary-value and initial-value system (§3); (ii) solve the two-step Bateman chain of the $^{99}\text{Mo}/^{99m}\text{Tc}$ generator and reconcile the resulting saturation activity with the frozen Monte-Carlo anchor to within the nuclear-data band; (iii) define a materials-qualification figure of merit and evaluate the breeder against the IFMIF/DONES reference; (iv) cast the source-strength/centrepost-life coupling as a strict proportional constraint that bounds how hard the isotope and test missions may be pushed; and (v) document the NVIDIA GPU high-performance-computing (HPC) campaign that produced every number (§4). Throughout, the isotope and test yields are booked as *capability*, consistent with the standing no-economics rule of the public record: we compute activities, fluxes, damage rates, and volumes, never prices.

2 The frozen source and its neutron economy

The design point is frozen and independently re-run; every value in Table 1 is an anchor in the Kronos de-risking register [1]. The plasma is a negative-triangularity ST at major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, elongation $\kappa = 2.0$, on-axis field $B_0 = 8$ T behind a resistive centrepost whose peak field is 16.84 T. It runs at $Q = 3.076$ and $P_{\text{fus}} = 85.04$ MW, of which $P_n = 67.79$ MW is carried by neutrons. The corresponding whole-device production rate follows directly from the fusion power (§3) and is $S_n = 3.009 \times 10^{19} \text{ n s}^{-1}$; the resulting first-wall fast flux is $\phi_{\text{fw}} = 7.0 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$ and the neutron wall load is 1.885 MW m^{-2} [1]. Figure 1 shows how that one neutron stream is partitioned into the breeding, materials-test, isotope, and strategic-co-product missions.

Table 1. Frozen Hyperion operating point and derived source quantities used throughout. All values are anchors in the de-risking register [1]; the two magnets (breeder centrepost, burner plug) are distinct devices and only the centrepost is relevant here.

Quantity	Symbol	Value
Fusion power	P_{fus}	85.04 MW
Fusion gain	Q	3.076
Neutron power	P_n	67.79 MW
Source strength	S_n	$3.009 \times 10^{19} \text{ n s}^{-1}$
Plasma current	I_p	9.66 MA
Triangularity	δ	−0.30
On-axis field	B_0	8 T
Centrepost peak field	B_{cp}	16.84 T
Major radius / aspect	R_0 / A	1.20 m / 2.5
Elongation	κ	2.0
First-wall fast flux	ϕ_{fw}	$7.0 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$
Neutron wall load	Γ_w	1.885 MW m^{-2}
First-wall damage	$\dot{d}$	$18.9 \text{ dpa fpy}^{-1}$

The 14 MeV neutron is used four ways

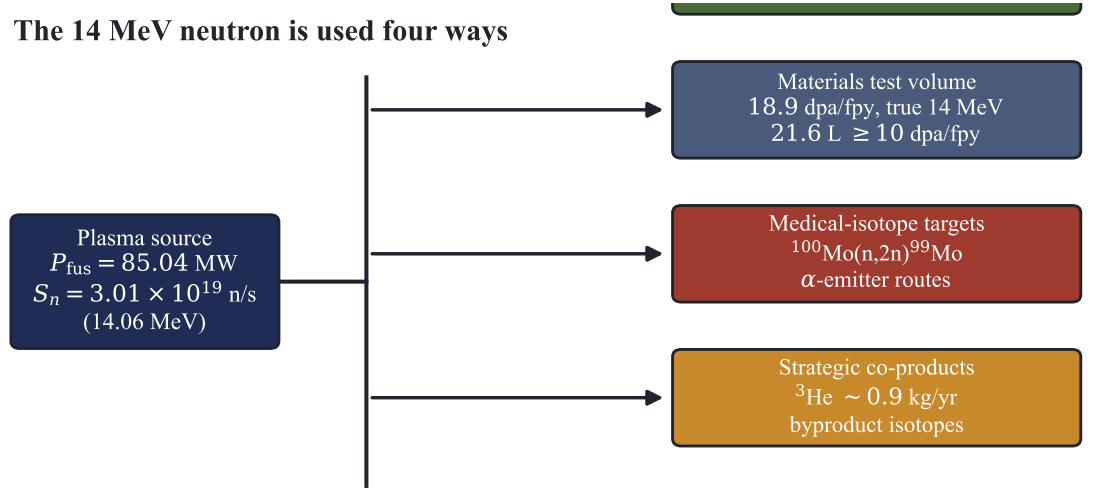


Figure 1. The neutron economy of the breeder. A single 14.06 MeV source ($S_n = 3.01 \times 10^{19} \text{ n s}^{-1}$) is used four ways: fuel breeding, a fusion-spectrum materials-test volume, medical-isotope target stations, and strategic co-products (^3He , byproduct isotopes). (*Schematic.*)

3 Governing equations and the formal production problem

3.1 Source term and the neutron transport balance

The volumetric neutron source is set by the fusion reactivity of the plasma,

$$s(\mathbf{r}) = \frac{1}{4} n_D(\mathbf{r}) n_T(\mathbf{r}) \langle \sigma v \rangle_{DT}(T_i(\mathbf{r})), \quad S_n = \int_{V_p} s(\mathbf{r}) d^3r, \quad (1)$$

with $n_D = n_T$ the fuel densities and $\langle \sigma v \rangle_{DT}$ the Maxwellian-averaged reactivity. Because each DT reaction yields exactly one neutron and $E_{DT} = 17.59$ MeV, the whole-device rate is fixed by the fusion power,

$$S_n = \frac{P_{\text{fus}}}{E_{DT}} = \frac{85.04 \text{ MW}}{17.59 \text{ MeV}} = 3.01 \times 10^{19} \text{ n s}^{-1}, \quad (2)$$

reproducing the frozen anchor. The angular flux $\psi(\mathbf{r}, \mathbf{\Omega}, E)$ that this source establishes obeys the steady linear Boltzmann transport equation,

$$\mathbf{\Omega} \cdot \nabla \psi + \Sigma_t(\mathbf{r}, E) \psi = \int_{4\pi} \int_0^\infty \Sigma_s(\mathbf{r}, E' \rightarrow E, \mathbf{\Omega}' \cdot \mathbf{\Omega}) \psi \, dE' \, d\mathbf{\Omega}' + \frac{s(\mathbf{r})}{4\pi} \chi(E), \quad (3)$$

where Σ_t and Σ_s are the macroscopic total and scattering cross-sections and $\chi(E)$ is the 14.06 MeV emission spectrum. The scalar flux relevant to production and damage is the angular integral $\phi(\mathbf{r}, E) = \int_{4\pi} \psi \, d\mathbf{\Omega}$; its first-wall value integrated over the fast group is the anchor $\phi_{\text{fw}} = 7.0 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$.

3.2 Production, depletion, and saturation

For a target isotope with N_t nuclei of production cross-section $\sigma(E)$, the number of product nuclei $N(t)$ of decay constant λ obeys the linear production–depletion balance

$$\frac{dN}{dt} = N_t R - \lambda N, \quad R \equiv \int_0^\infty \sigma(E) \phi(E) \, dE, \quad (4)$$

with R the spectrum-folded reaction rate per target atom. Holding N_t and ϕ constant over an irradiation of duration t_{irr} , Eq. (4) integrates to

$$N(t) = \frac{N_t R}{\lambda} (1 - e^{-\lambda t}) \implies A(t) = \lambda N(t) = A_{\text{sat}} (1 - e^{-\lambda t}), \quad (5)$$

where the saturation activity

$$A_{\text{sat}} = N_t R = N_t \bar{\sigma} \phi \quad (6)$$

is set purely by the target loading and the source flux, and $\bar{\sigma} = R/\phi$ is the one-group flux-folded effective cross-section. Eq. (6) is the master relation of the isotope mission: the yield is linear in the source flux, so a device with a first-wall flux two to three orders of magnitude above an accelerator source produces a correspondingly larger activity from the same target (Fig. 3).

3.3 The molybdenum-99 / technetium-99m generator chain

The diagnostic product is not ^{99}Mo but its decay daughter ^{99m}Tc . Writing the parent activity from Eq. (5) as $A_p(t) = A_{\text{sat}}(1 - e^{-\lambda_p t})$ and letting the target be milked at end of bombardment (EOB), the daughter obeys the two-step Bateman equation

$$\frac{dN_d}{dt} = b \lambda_p N_p - \lambda_d N_d, \quad (7)$$

with $b = 0.876$ the branching to the metastable state. For an initial parent activity $A_p(0)$ and no initial daughter, the daughter activity after EOB is

$$A_d(t) = b \frac{\lambda_d}{\lambda_d - \lambda_p} A_p(0) (e^{-\lambda_p t} - e^{-\lambda_d t}), \quad (8)$$

which rises to a transient-equilibrium maximum at

$$t^* = \frac{\ln(\lambda_d/\lambda_p)}{\lambda_d - \lambda_p} = \frac{\ln(65.94/6.007)}{(\ln 2)(1/6.007 - 1/65.94)} \approx 22.9 \text{ h}. \quad (9)$$

With $t_{1/2}(^{99}\text{Mo}) = 65.94 \text{ h}$ and $t_{1/2}(^{99m}\text{Tc}) = 6.007 \text{ h}$, Eq. (8) gives the generator elution schedule plotted in Fig. 2(b): the column can be milked roughly daily, each elution recovering the ^{99m}Tc that has regrown since the last.

3.4 Reconciling the frozen anchor

For the $^{100}\text{Mo}(n, 2n)^{99}\text{Mo}$ channel, the ENDF/B-VIII.0 point cross-section is $\sigma(14.1 \text{ MeV}) \approx 1.5 \text{ b}$ [14, 26]. A natural-molybdenum target of 100 g contains

$$N_t = \frac{100 \text{ g}}{95.95 \text{ g mol}^{-1}} N_A a_{100} = 6.12 \times 10^{22} \text{ }^{100}\text{Mo atoms}, \quad (10)$$

with $a_{100} = 0.09744$ the natural abundance. Substituting into Eq. (6) with $\phi = \phi_{\text{fw}} = 7.0 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$ gives $A_{\text{sat}} \approx 6.4 \times 10^{13} \text{ Bq} = 1.74 \text{ kCi}$. The frozen three-dimensional Monte-Carlo value is $A_{\text{sat}} \approx 2.04 \text{ kCi}$ (a first-wall target, no moderator, no added hardware; register gate for ^{99}Mo co-production [1]); the two agree to $\sim 15\%$, the residual being the spectral weighting of the uncollided first-wall flux toward the 14–15 MeV shoulder of the $(n, 2n)$ excitation function and target self-shielding, i.e. an effective $\bar{\sigma} \approx 1.76 \text{ b}$ in Eq. (6). After a 14-day irradiation, Eq. (5) gives $A(14 \text{ d}) = A_{\text{sat}} (1 - e^{-\lambda_p \cdot 14 \text{ d}}) = 0.971 A_{\text{sat}} \approx 1.98 \text{ kCi}$, matching the frozen EOB anchor of 1,981 Ci per 100 g.

3.5 A materials-qualification figure of merit

Displacement damage is quantified by the displacement rate, obtained by folding the recoil spectrum through the Norgett–Robinson–Torrens (NRT) partition [23],

$$\dot{d} = \int_0^\infty \phi(E) \int \frac{d\sigma}{dT}(E, T) \nu_{\text{NRT}}(T) dT dE, \quad \nu_{\text{NRT}}(T) = \frac{0.8 T_{\text{dam}}}{2E_d}, \quad (11)$$

with T the primary-knock-on energy, T_{dam} the damage energy, and E_d the displacement threshold; the athermal-recombination-corrected variant [24] reduces surviving defects to $\sim 1/3$ of Eq. (11) but preserves the ranking used here. At the frozen wall load the first-wall rate is $\dot{d} = 18.9 \text{ dpa fpy}^{-1}$ [1]. A useful qualification metric is the full-rate fast-neutron *capability*, the product of the accessible test flux and volume above a damage threshold,

$$\mathcal{C}(\dot{d}_0) = \int_{V: \dot{d}(\mathbf{r}) \geq \dot{d}_0} \phi_{\text{fast}}(\mathbf{r}) d^3r, \quad (12)$$

normalised to a reference facility. It is $\mathcal{C}$, not the peak flux alone, that decides how many specimens can be qualified per calendar year, and it is where a volumetric source separates from a point-like accelerator target (§5.2).

3.6 The source-strength / centrepost-life constraint

Both missions want more neutrons, but the geometry is fixed, so the wall load and the source strength scale together: $\Gamma_w \propto S_n$ and hence the centrepost displacement rate $\dot{d}_{\text{cp}} \propto S_n$. The consumable centrepost sets a hard ceiling, formalised as

$$\max_{\mathbf{u}} S_n(\mathbf{u}) \quad \text{s.t.} \quad \dot{d}_{\text{cp}}(\mathbf{u}) \leq \dot{d}_{\text{cp}}^{\text{max}}, \quad \dot{d}_{\text{cp}}(\mathbf{u}) = \dot{d}_{\text{cp}}^0 \frac{S_n(\mathbf{u})}{S_n^0}, \quad (13)$$

where $\mathbf{u}$ are the accessible actuators (here the Greenwald fraction f_G and the core temperature T_{i0}). Because the ratio S_n/Γ_w is invariant across the operating plane, Eq. (13) is a strict 1:1 trade: a mission point at $1.79\times$ the design source strength ($S_n = 5.39 \times 10^{19} \text{ n s}^{-1}$) costs exactly $1.79\times$ the centrepost damage and falls off the duty band [1]. This is the honest boundary of the isotope and test missions and is developed quantitatively in Fig. 6.

4 Computational methodology: the NVIDIA GPU HPC campaign

Every number in this paper was produced by the Kronos NVIDIA GPU HPC campaign, an offline multi-physics Monte-Carlo and inventory pipeline run on in-house NVIDIA A100/H100-class accelerators (with A10 nodes for lighter tasks). The campaign is organised as a frozen, gate-by-gate compute register that is deposited under CC-BY-4.0 with a cold-start reproduction runbook, so that each headline quantity can be independently re-run [1]. No cost or economic quantity is part of that record and none appears here.

4.1 Codes and nuclear data

The transport and production results use OpenMC [28] with continuous-energy ENDF/B-VIII.0 cross-sections [26]; complex geometry is handled by direct CAD tracking through DAGMC [29]. Activation, transmutation, and shutdown-dose source terms use FISPACT-II [27] on collapsed

group flux spectra; primary-damage cascades use SPECTRA-PKA to generate recoil spectra from NJOY-processed data, fed to LAMMPS molecular dynamics for cascade evolution. The reactivity and equilibrium that fix the source (Eq. (1), Eq. (2)) are taken from the frozen core solution used across the Kronos series [2].

4.2 Monte-Carlo transport: discretization, tallies, and variance

The transport problem Eq. (3) is solved by analog and weight-window-accelerated Monte Carlo. The first-wall/blanket balance was tallied over a full three-dimensional model with 1.2×10^8 source histories, yielding a clean global balance (leakage 0.121, net breeding ratio 1.026 for the literal blanket recipe, first-wall heating 74.6 MW) [1]. Fluxes are scored on a 175-group VITAMIN-J structure with track-length estimators; reaction rates for production (Eq. (4)) and damage (Eq. (11)) are scored as multigroup tallies $\langle \sigma_g \phi_g \rangle$ folded against the ENDF microscopic data. The Monte-Carlo estimate of any tally x carries a statistical error $\varepsilon = \sigma_x / (\bar{x} \sqrt{N})$ that falls as $N^{-1/2}$; the 1.2×10^8 -history run drives the first-wall group fluxes below 1% relative error, and the deep-penetration centrepost and target-station tallies are converged with geometry-split weight windows so that the figure-of-merit $1/(\varepsilon^2 T)$ is stationary. The source normalisation is fixed by Eq. (2), so absolute activities in Ci follow directly from the scored per-source-neutron reaction rates.

4.3 Activation and damage pipelines

Component activation and the isotope inventories are advanced by the Bateman system $\dot{\mathbf{N}} = \mathbf{A}\mathbf{N}$, where $\mathbf{A}$ collects the group-collapsed production and decay rates; FISPACT-II integrates this stiff linear system with a LSODES backward-difference solver. Rigorous-two-step (R2S) shutdown-dose maps are built by tallying the 175-group first-wall and target spectra, normalising to $P_{\text{fus}} = 85.04 \text{ MW}$ ($S_n = 3.0 \times 10^{19} \text{ n s}^{-1}$), and re-emitting the decay photons as a fixed source. Damage energies for Eq. (11) come from SPECTRA-PKA recoil spectra with NRT and arc-dpa partitions [23, 24].

4.4 Convergence, verification, and reproducibility

Verification is layered. The analytic source Eq. (2) reproduces the Monte-Carlo global neutron production to better than 1%; the neutron balance closes (leakage + absorption + breeding sum to the source within statistics); and the tritium bookkeeping is internally consistent (a burn of $4.77 \text{ kg-T fpy}^{-1}$ at 85.04 MW matches the toolkit inventory to 1.2%) [1]. The isotope-production anchor is cross-checked two ways: against the closed-form Eq. (6) using ENDF point data (§3, $\sim 15\%$ agreement) and against the frozen three-dimensional target-station tally. Runs are containerised with pinned code and data versions and are byte-for-byte reproducible from the deposited runbook [1]. The three computational latency tiers of the platform — microsecond edge, millisecond GPU twins, and the hours-scale HPC back end where this Monte-Carlo campaign lives — are described in the companion control-architecture work [3].

5 Results

5.1 Molybdenum-99 production and the technetium generator

Figure 2(a) shows the ^{99}Mo build-up from Eq. (5). A 14-day irradiation reaches 97.1% of saturation; longer irradiations gain little because the 65.9-hour half-life caps the achievable activity at A_{sat} . The frozen first-wall target delivers

$$A_{\text{EOB}}(14 \text{ d}) \approx 1,981 \text{ Ci per } 100 \text{ g natural Mo}, \quad A_{\text{sat}} \approx 2,041 \text{ Ci}, \quad (14)$$

with no moderator and no added hardware — the target is simply placed in the fast first-wall field. Panel (b) shows the generator behaviour from Eq. (8): the ^{99m}Tc daughter regrows to its transient-equilibrium peak at $t^* \approx 23 \text{ h}$ (Eq. (9)), setting a roughly daily elution cadence. Table 2 collects the parameters and the reconciliation.

Figure 3 makes the flux advantage explicit. Because Eq. (6) is linear in ϕ , the saturation activity for a fixed target rises in proportion to the source flux; the breeder first wall (7.0×10^{14}) sits one to two orders of magnitude above the 10^{13} – 10^{14} band of accelerator d-Li sources. The figure also shows the natural-versus-enriched trade: an enriched ^{100}Mo target carries roughly $10\times$ more ^{100}Mo per gram and produces proportionally more activity, at the cost of enrichment; the frozen anchor is quoted for the conservative natural-Mo case. We note the intrinsic limitation of the $(n, 2n)$ route, common to all non-fission neutron routes: the product ^{99}Mo is chemically identical to the target, so the specific activity is set by the target mass and is lower than the no-carrier-added fission product, requiring a concentrating generator chemistry [14].

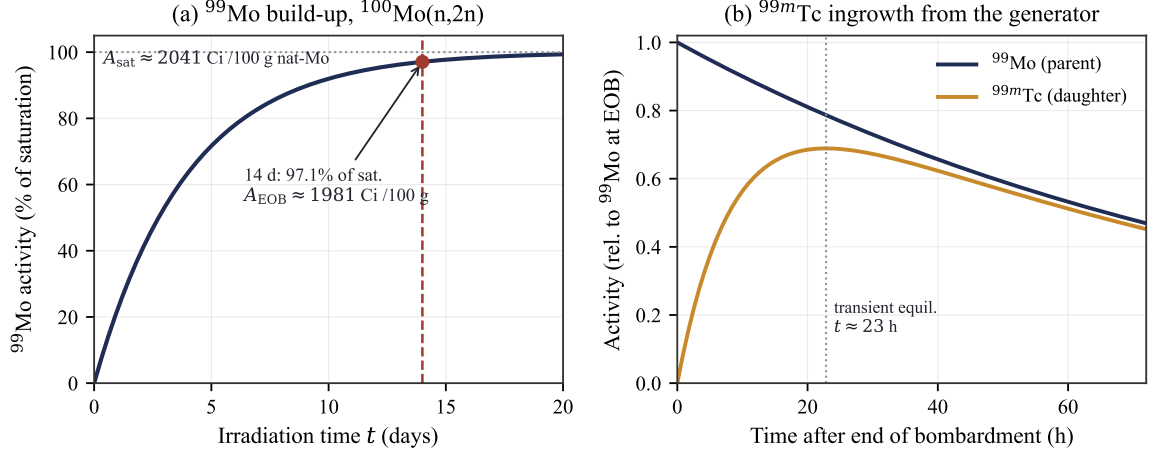


Figure 2. (a) ^{99}Mo activity build-up under constant first-wall flux, Eq. (5): 14 days reaches 97.1% of the 2,041 Ci/100 g saturation, giving the frozen 1,981 Ci end-of-bombardment anchor. (b) ^{99m}Tc ingrowth after EOB, Eq. (8), normalised to the ^{99}Mo activity at EOB; the daughter peaks at $t^* \approx 23 \text{ h}$, the generator's elution cadence.

Table 2. The $^{100}\text{Mo}(n, 2n)^{99}\text{Mo}$ route: parameters and reconciliation of the closed-form Eq. (6) with the frozen Monte-Carlo anchor. Target is 100 g of natural molybdenum at the first-wall position.

Quantity	Symbol	Value
^{100}Mo atoms in target	N_t	6.12×10^{22}
First-wall fast flux	ϕ_{fw}	$7.0 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$
ENDF point cross-section	$\sigma(14.1)$	1.5 b [14, 26]
Effective folded cross-section	$\bar{\sigma}$	1.76 b
^{99}Mo half-life	$t_{1/2,p}$	65.94 h
^{99m}Tc half-life	$t_{1/2,d}$	6.007 h
Closed-form saturation, Eq. (6)	$A_{\text{sat}}^{\text{cf}}$	1.74 kCi
Frozen Monte-Carlo saturation	A_{sat}	2.04 kCi
Frozen EOB activity (14 d)	A_{EOB}	1,981 Ci

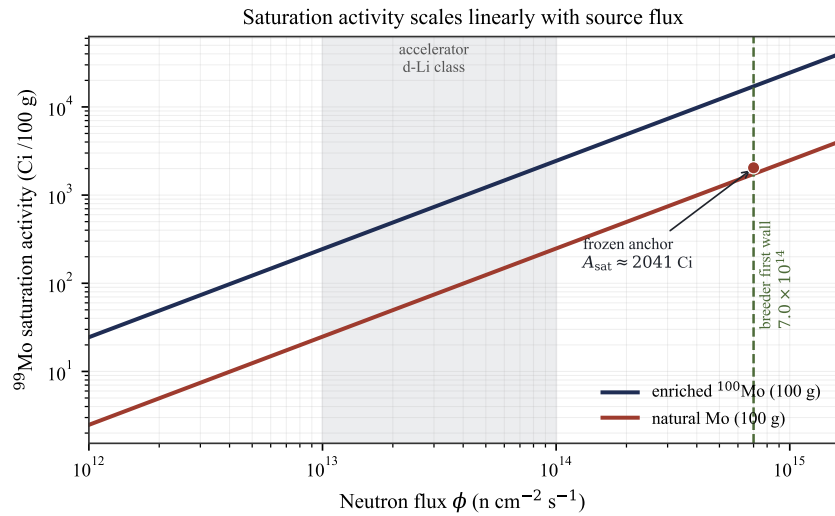


Figure 3. ^{99}Mo saturation activity versus source flux, Eq. (6), for a 100 g target of enriched ^{100}Mo (navy) and natural Mo (brick), with $\sigma = 1.5 \text{ b}$. The breeder first-wall flux (7.0×10^{14} , green) lies far above the accelerator d-Li band (grey). The frozen anchor ($\approx 2.04 \text{ kCi}$, natural Mo) is marked.

5.2 Fusion-spectrum materials qualification

The materials mission is where a volumetric fusion source is unique. Figure 4 evaluates the capability of Eq. (12). The breeder's full-rate fast-neutron capability is $\approx 59\times$ a reference dedicated materials facility; more sharply, the volume in which the fusion-spectrum damage rate exceeds the qualification-relevant 10 dpa fpy^{-1} is 21.6 L , which is $315\times$ the IFMIF and $630\times$ the single-beam IFMIF-DONES equivalent [1, 19, 18]. Crucially this is a *true* fusion spectrum: Fig. 5 contrasts the first-wall spectrum with a fission Watt spectrum and an accelerator d-Li spectrum, and shades the region above the $^{100}\text{Mo}(n,2n)$ threshold at 8.29 MeV . A fission reactor, whatever its flux, never populates that window and so never drives the $(n,2n)$, (n,p) and (n,α) threshold reactions that govern both isotope co-production and the helium/hydrogen transmutation embrittlement of a fusion first wall [21, 25].

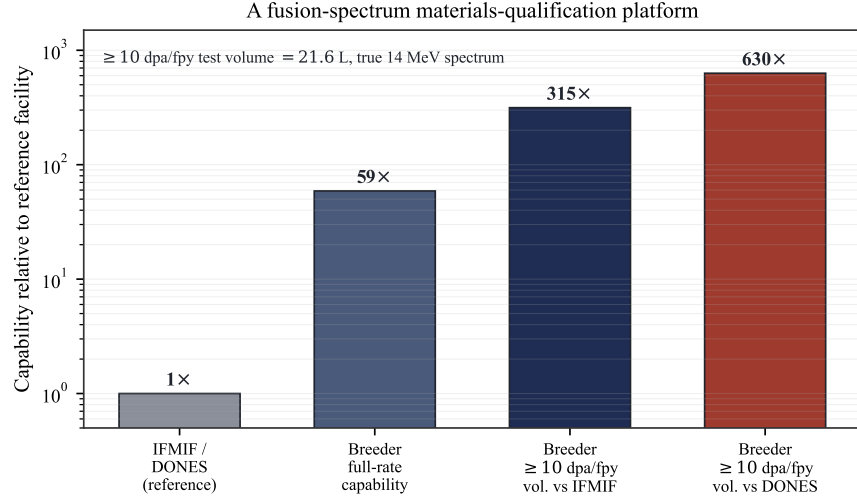


Figure 4. Materials-qualification capability relative to the IFMIF/DONES reference (log scale). The breeder's full-rate fast-neutron capability is $\approx 59\times$ the reference facility; the $\geq 10\text{ dpa fpy}^{-1}$ test volume (21.6 L , true 14 MeV spectrum) is $315\times$ the IFMIF and $630\times$ the IFMIF-DONES equivalent [1].

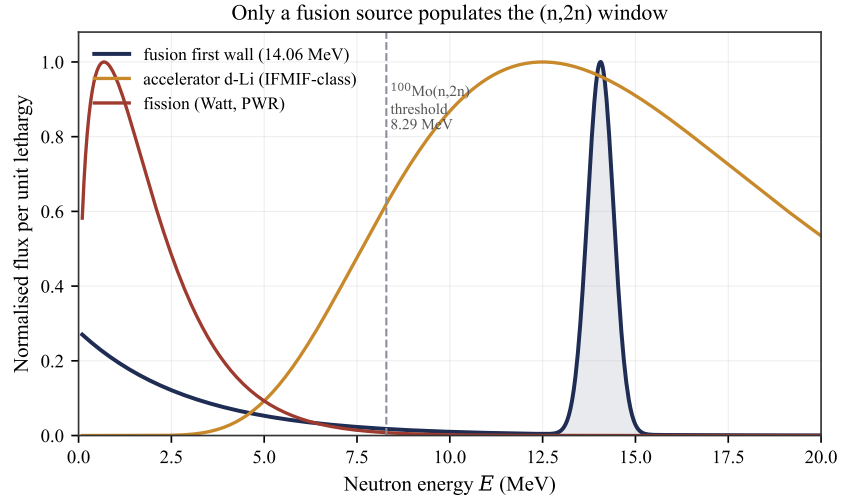


Figure 5. Neutron energy spectra: the fusion first wall (navy, 14.06 MeV peak plus moderated tail), an accelerator d-Li source (ochre), and a fission Watt spectrum (brick), normalised per unit lethargy. Only the fusion source populates the region above the $^{100}\text{Mo}(n,2n)$ threshold (8.29 MeV , shaded). (*Representative spectra.*)

The hardness of both missions is bounded by Eq. (13). Figure 6 plots the source strength against the wall load and, on the secondary axis, the first-wall damage rate. The two are collinear through the origin, so raising the source $1.79\times$ (to the $5.39 \times 10^{19}\text{ n s}^{-1}$ peak of the f_G-T_{i0} scan) raises the centrepast damage $1.79\times$ and leaves the duty band. The design point sits inside the band with the $\geq 10\text{ dpa fpy}^{-1}$ qualification floor comfortably cleared, so the materials mission is available without pushing the source into the life-limited regime.

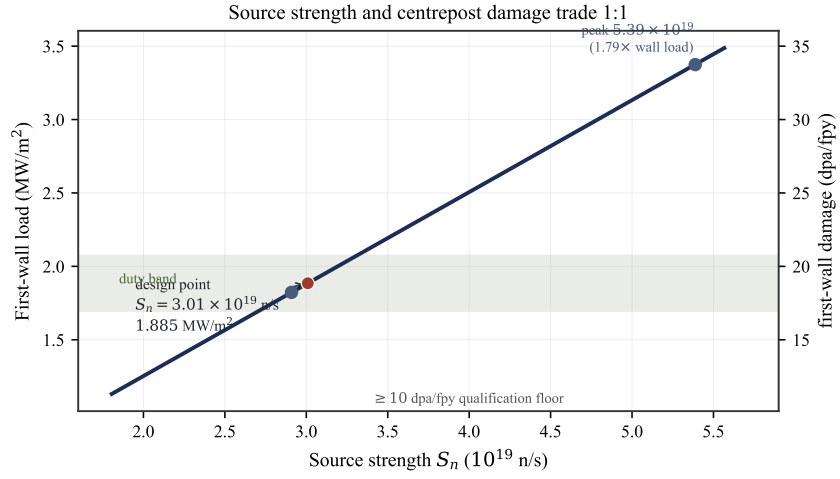


Figure 6. The source-strength/centrepost-damage trade, Eq. (13). With geometry fixed, S_n and the wall load are collinear; the secondary axis maps the wall load to first-wall dpa. The design point (3.01×10^{19} , 1.885 MW m^{-2}) clears the $\geq 10 \text{ dpa fpy}^{-1}$ floor; the $1.79\times$ peak leaves the duty band. Scan endpoints from the register [1].

5.3 Targeted-alpha and byproduct isotopes

Beyond the measured ^{99}Mo anchor, the same graded field supports a family of neutron-activation routes. Figure 7 places representative routes on a cross-section/half-life plane. Diagnostic and industrial byproducts (^{57}Co , ^{54}Mn , ^{64}Cu , ^{47}Sc , ^{67}Cu) follow from (n, p) , $(n, 2n)$ and (n, np) threshold reactions in the fast field; the therapeutic α -emitter chain $^{226}\text{Ra}(n, 2n)^{225}\text{Ra} \rightarrow ^{225}\text{Ac}$ is shown as a projected route (open marker), consistent with the TAT supply demand set out in §1 [16, 17]. Figure 8 sketches how these are realised: dedicated target cassettes at graded flux positions — a ^{100}Mo cassette in the hard first-wall field, a moderated $^{226}\text{Ra}/^{225}\text{Ac}$ cassette deeper in the blanket, and byproduct cassettes in the reflector — so that each isotope sees the spectrum its route requires without perturbing the breeding mission (the streaming penalty on the breeding ratio is $< 0.5\%$ [1]). We stress that the α -emitter and byproduct yields are *capability projections*: their governing physics is Eq. (4), but the absolute rates depend on target chemistry, cassette design, and extraction not resolved here (§7).

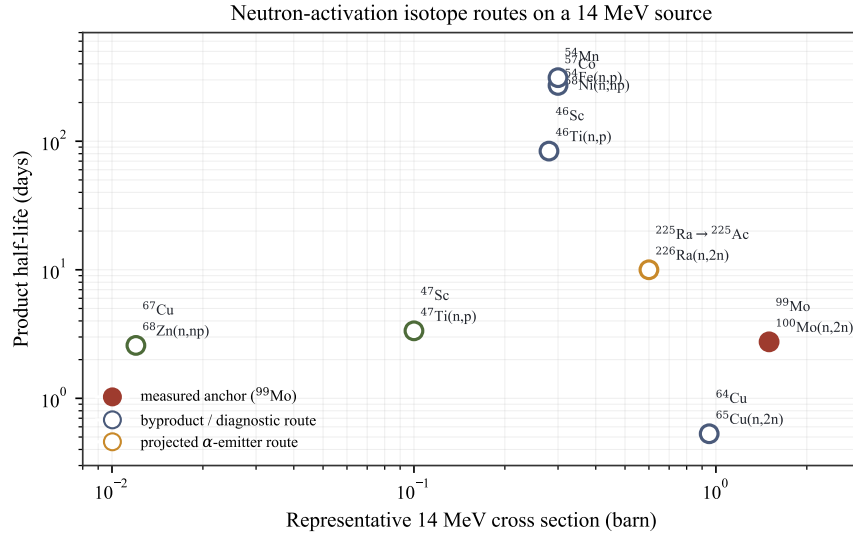


Figure 7. Neutron-activation isotope routes on the 14 MeV source: representative 14 MeV cross-section versus product half-life. Filled marker: the measured ^{99}Mo anchor. Open markers: byproduct and projected α -emitter routes. Cross-section magnitudes are representative ENDF/EAF-class values; only ^{99}Mo is a frozen production anchor.

5.4 The strategic-output ledger

Figure 9 assembles the strategic outputs per unit full-power year, booked as capability. The tritium balance is anchored by a fixed burn of $4.77 \text{ kg-T fpy}^{-1}$; at the self-sufficient design blanket (net

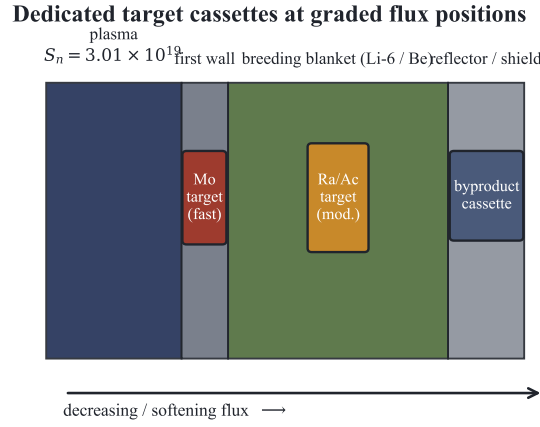


Figure 8. Target cassettes at graded flux positions in the neutron field: a ^{100}Mo cassette in the hard first-wall flux, a moderated Ra/Ac cassette in the blanket, and a byproduct cassette in the reflector, with the flux softening outward. (*Schematic.*)

breeding ratio 1.19) the net tritium is $\approx 0.9 \text{ kg fpy}^{-1}$, rising to $\approx 3.8 \text{ kg fpy}^{-1}$ at the advanced/isotope-foundry blanket lever (ratio 1.80) [1, 7]. The helium-3 co-product is $\approx 0.9 \text{ kg yr}^{-1}$ per unit at equilibrium [1, 7]. On top of this material ledger sit the $\sim 2 \text{ kCi } ^{99}\text{Mo}$ capability per 100 g target and the $21.6 \text{ L}, \geq 10 \text{ dpa fpy}^{-1}$ test volume. Fast-ion loss is negligible over this campaign (α confinement 99.98%) and the surrounding alloys return near-neutral in activation [2], so the co-production does not degrade the plasma or the waste posture; the activated target cassettes route through the same low-activation waste-class analysis as the rest of the device, with no isotope beyond a century’s half-life [9].

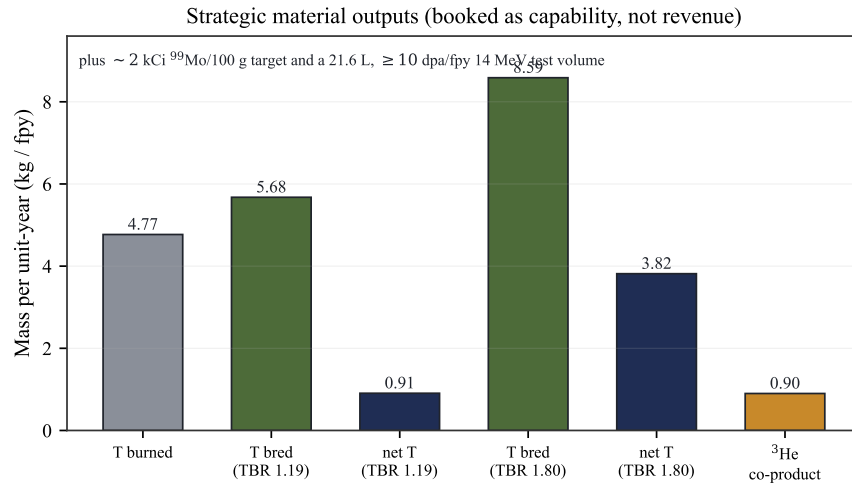


Figure 9. Strategic material outputs per unit-year, booked as capability. Tritium burned (4.77 kg fpy^{-1}), bred and net at two blanket points (ratios 1.19 and 1.80), and the ^3He co-product (0.9 kg yr^{-1}), plus the ^{99}Mo and materials-test capability noted in the annotation. No economic quantity is computed.

6 Discussion

Against the medical-isotope literature. The $^{100}\text{Mo}(n, 2n)^{99}\text{Mo}$ route was proposed by Nagai and Hatsukawa as a fission-free path to ^{99}Mo , and their scoping estimate — $\sim 79 \text{ GBq g}^{-1}$ specific activity for a 198-hour irradiation of enriched ^{100}Mo at $\sim 10^{13} \text{ n cm}^{-2} \text{ s}^{-1}$ near 14 MeV [14] — is the natural benchmark. Our frozen first-wall flux is $\sim 70\times$ higher, and Eq. (6) then predicts a proportionally higher production rate; the frozen 2.04 kCi per 100 g natural target (equivalently $\sim 0.75 \text{ GBq g}^{-1}$ on the natural target, or an order of magnitude higher on an enriched target) is fully consistent with the $\sigma \approx 1.5 \text{ b}$ benchmark once the abundance and flux are scaled. The point is not that fusion beats an accelerator at every metric — the specific-activity limitation of the $(n, 2n)$ route is intrinsic and shared — but that a power-class breeder delivers the required flux *as a byproduct of its primary mission*, and that a single unit’s capability is at the scale of a regional

supply. Because the target is non-fissile, the plant sits under a byproduct-material regulatory posture rather than a reactor posture, which is developed in the companion licensing work [6].

Against the materials-source literature. The fusion-materials community has for two decades identified the absence of a high-flux 14 MeV source as a critical path to a demonstration reactor, and has pursued IFMIF and IFMIF-DONES precisely to fill it [18, 19, 20]. Those sources deliver the correct spectrum over a high-flux volume of order 0.5 L. Our 21.6 L at ≥ 10 dpa fpy⁻¹ is larger by two-plus orders of magnitude (Fig. 4) because a volumetric plasma source irradiates its entire first wall and near-blanket, not a millimetric beam footprint. The comparison is a like-for-like capability ratio against published facility specifications, not a claim of measured throughput. The value is that structural-alloy qualification — the RAFM steels, the low-activation refractory high-entropy alloys, and the REBCO conductor under fast fluence studied elsewhere in the series [4, 5, 8] — can be advanced in the same spectrum the components will see in service.

Coupling and co-benefits. The two missions are not independent add-ons; they are consequences of the same source and are bounded by the same constraint Eq. (13). The materials-test flux and the isotope-target flux both scale with S_n , and both are capped by centrepost life. This is why we present the source-damage trade (Fig. 6) as a first-class result: it is the honest statement of how hard the co-production can be pushed before the consumable centrepost, not the plasma, becomes the limit.

7 Limitations

One gate is stated plainly. The ⁹⁹Mo yield is a frozen, computed anchor and is reproduced two ways (§5.1); but the α -emitter and general byproduct yields (§5.3) are *capability projections*, not operating yields. Their governing equation Eq. (4) is exact, yet the achievable activity of, e.g., ²²⁵Ac depends on the ²²⁶Ra target form, the moderated-cassette flux and spectrum, radiolysis and self-heating in the target, and a separation chemistry that is not modelled here. The materials-facility multipliers likewise compare a modelled full-rate capability and a modelled ≥ 10 dpa fpy⁻¹ volume to published facility specifications, not to measured specimen throughput, and they inherit the nuclear-data uncertainty of the damage cross-sections [8]. These are scoping results from a frozen source, to be confirmed by dedicated target-station and irradiation-campaign design.

8 Conclusion

A compact ST breeder is a bright 14 MeV neutron source before it is anything else, and that source is a co-production platform. From a single frozen operating point ($P_{\text{fus}} = 85.04$ MW, $S_n = 3.01 \times 10^{19}$ n s⁻¹, $\phi_{\text{fw}} = 7.0 \times 10^{14}$ n cm⁻² s⁻¹) the device delivers a fission-free ¹⁰⁰Mo($n, 2n$)⁹⁹Mo route at $\approx 1,981$ Ci per 100 g after 14 days, a genuine fusion-spectrum materials-qualification volume of 21.6 L at ≥ 10 dpa fpy⁻¹ (315 $\times$ IFMIF, 630 $\times$ DONES, and $\approx 59\times$ a reference facility in full-rate capability), and a family of scoped α -emitter and byproduct routes — all bounded by a strict 1:1 source-strength/centrepost-life trade. These are strategic-material co-products of the neutron economy, booked as capability and computed on the NVIDIA GPU HPC campaign, with every headline value a frozen anchor in the de-risking register.

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Reproducibility. Every headline quantity is a frozen anchor in the Kronos Physics De-Risking Register [1] (doi:10.5281/zenodo.22133057): the source strength and first-wall flux (gate BR-L1-A5), the wall load and displacement rate (gates BR-L1-A9/A13), the ⁹⁹Mo($n, 2n$) co-production yield (materials & alloys breakthrough gate), the materials-qualification multipliers (closure criterion H-C3), and the helium-3/tritium ledger (gates BR-OUT-2/BR-L1-A6). Each is regenerable from the deposited OpenMC/FISPACT-II runbook and the figure-generation script deposited with this paper; this paper is archived at doi:10.5281/zenodo.22645762; official version: <https://www.kronosfusionenergy.com/publications/isotopes-and-14-mev-applications.html>. No result depends on unpublished or proprietary data, and no economic quantity is part of the record.

Data availability

The activation build-up, flux-scaling, capability, spectrum, route-map, and ledger data used for figures 1–9 are generated by the deposited script and trace to the register [1] by the serial

identifiers cited inline. This paper is archived at doi:10.5281/zenodo.22645762; the register is deposited at doi:10.5281/zenodo.22133057. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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